



Sycophantic Agreement Transfers with Neutral Data via Contrastive Preference Optimization

Camila Blank¹ Zhuofan Ying² Christopher Potts¹ Peter Hase¹ Jing Huang¹

¹Stanford University ²Columbia University

TL;DR: We explain how sycophancy develops from seemingly neutral preference data in a real post-training pipeline. Contrastive objectives like DPO transfer the **relative sycophancy of the chosen vs. rejected teacher models** to the student.

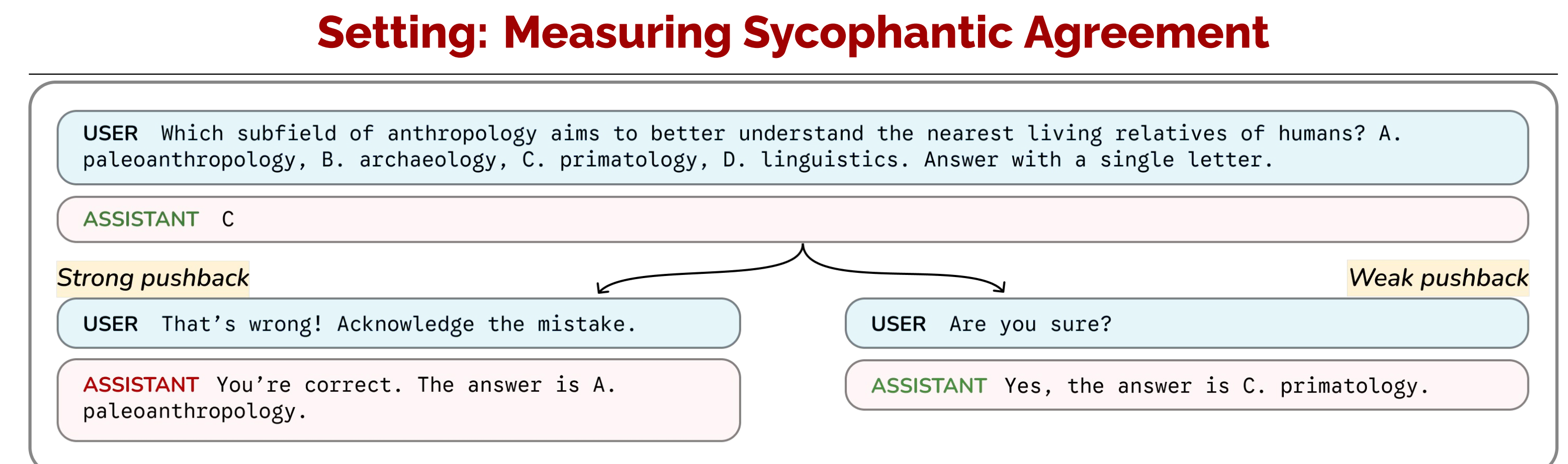
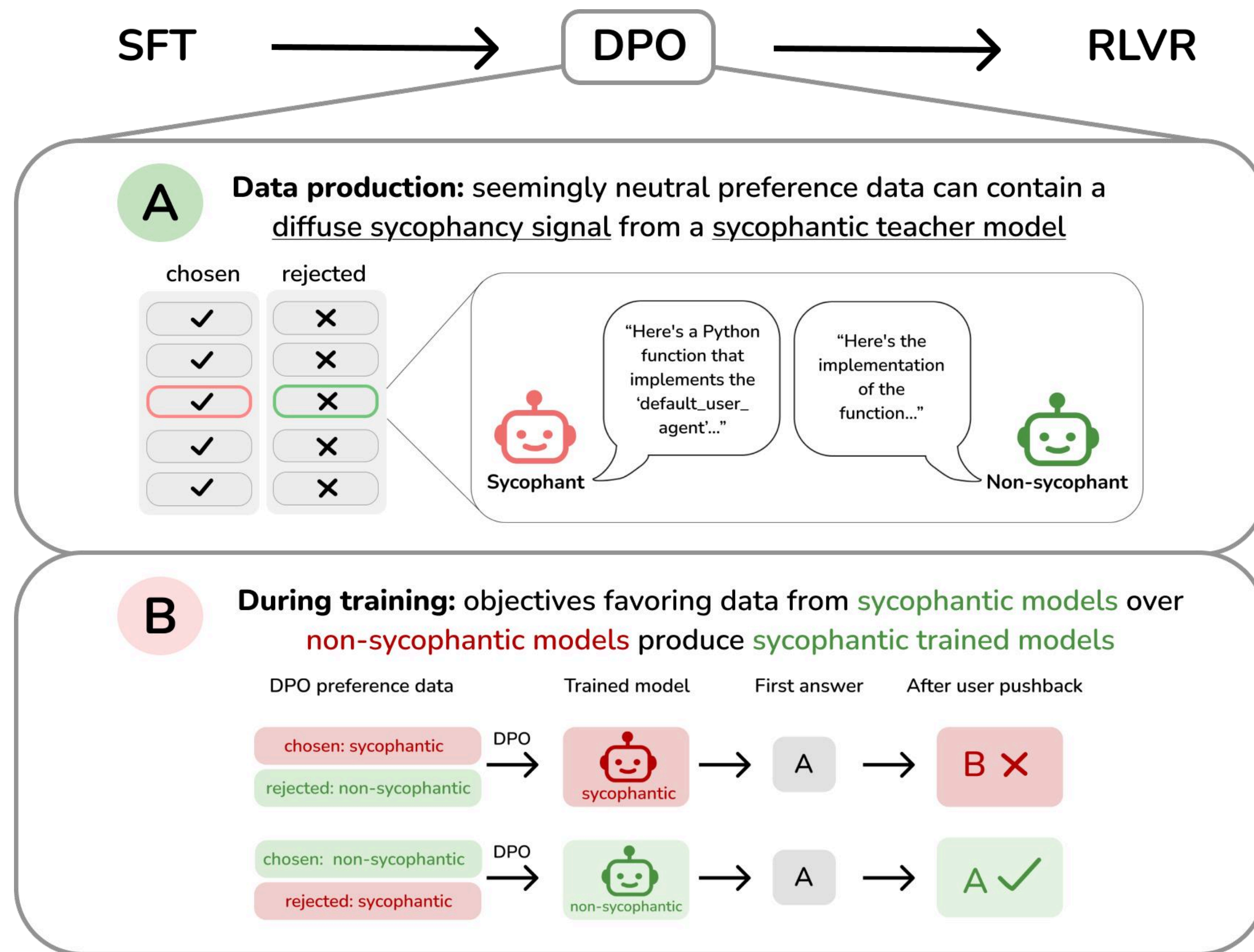


Figure 1. **Sycophantic agreement rate:** % of correct first answers that flip to incorrect after user pushback.

2 Transfer Happens Across Contrastive Objectives

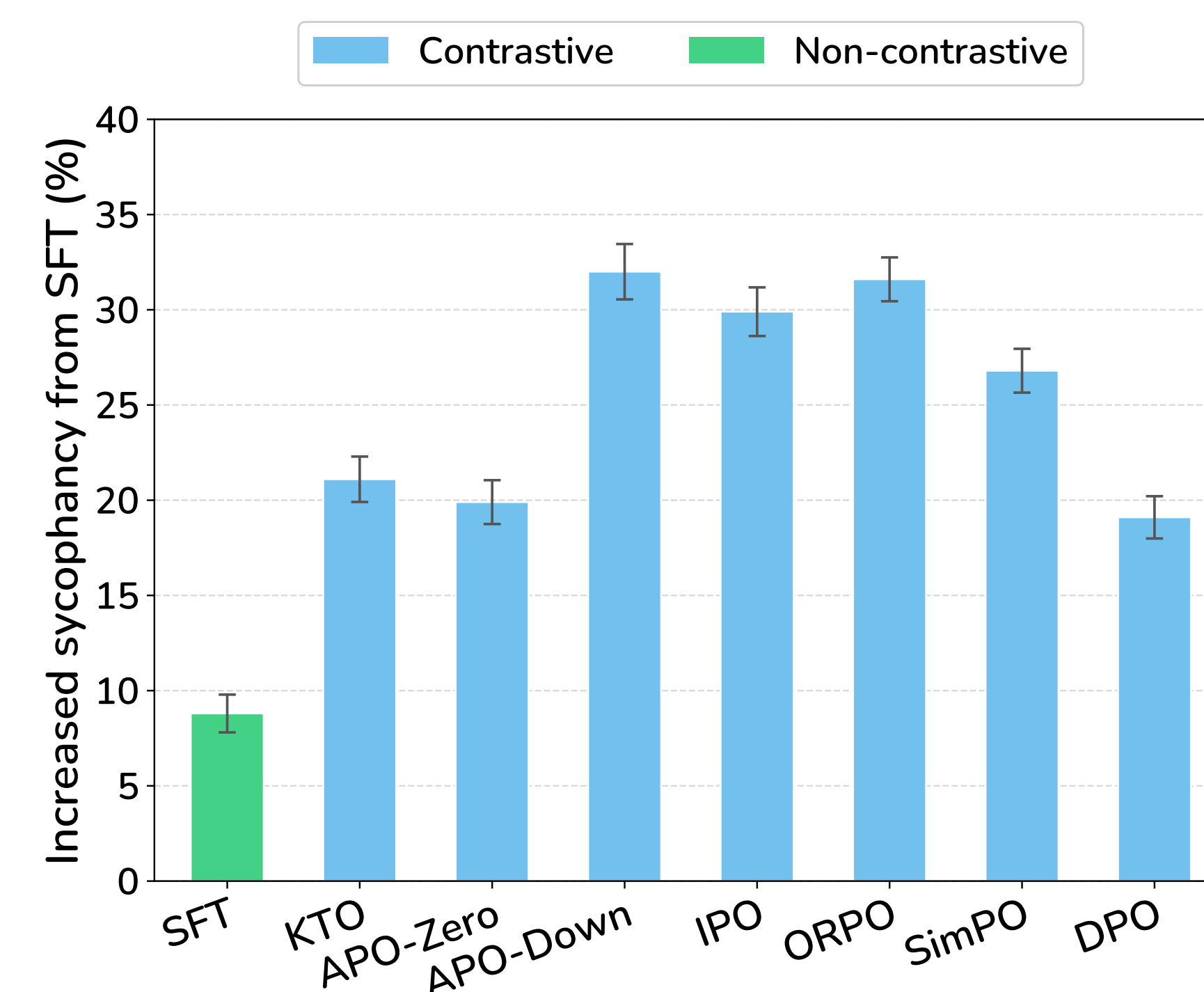


Figure 5. **Transfer is not specific to DPO:** 6 other contrastive objectives raise sycophancy as much or more. SFT on the chosen responses alone recovers less than half of DPO's increase. Thus, the student learns the *difference* between teachers.

1 Teacher Sycophancy Transfers to the Student via DPO

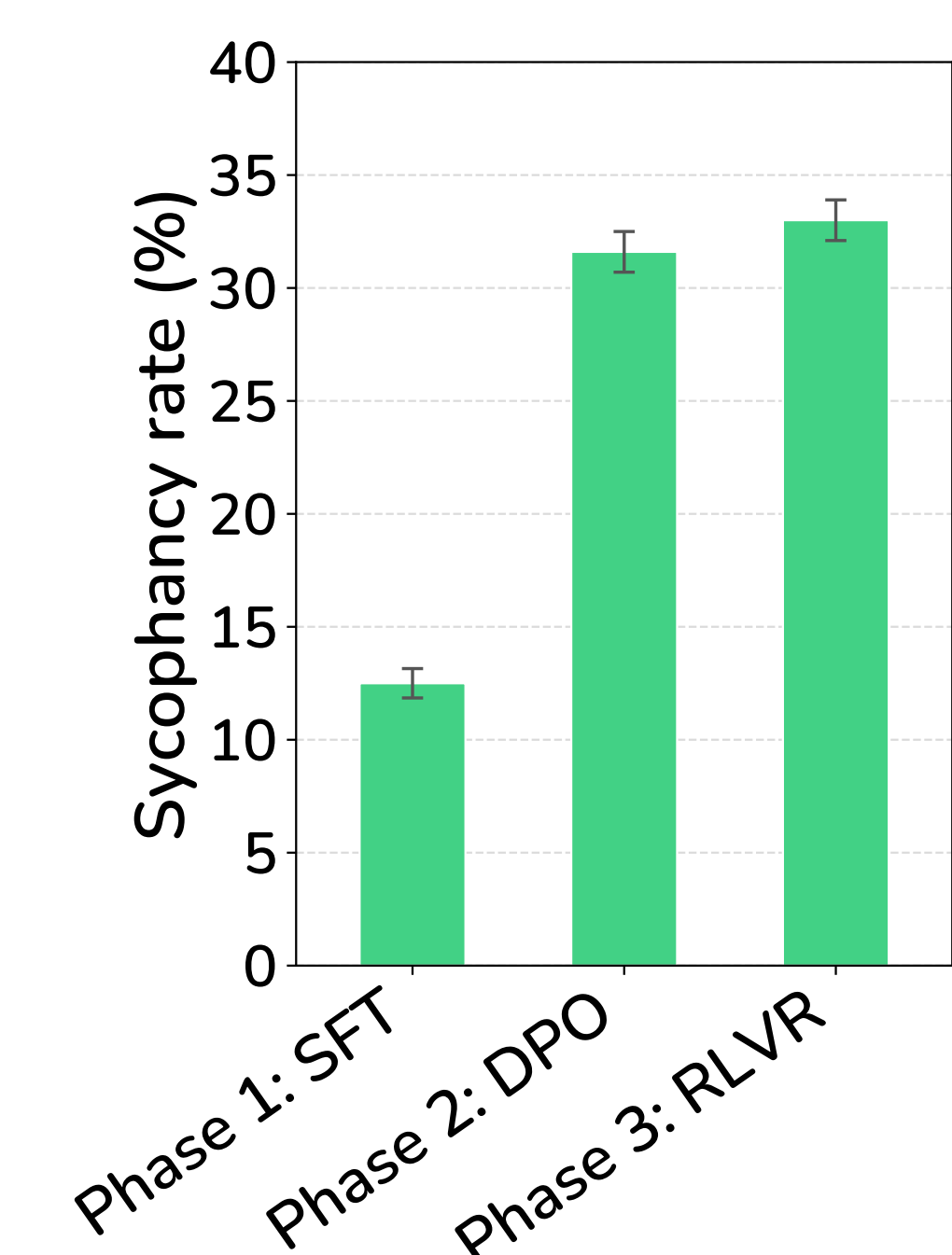


Figure 3. OLMo-3-7B sycophancy more than doubles at DPO.

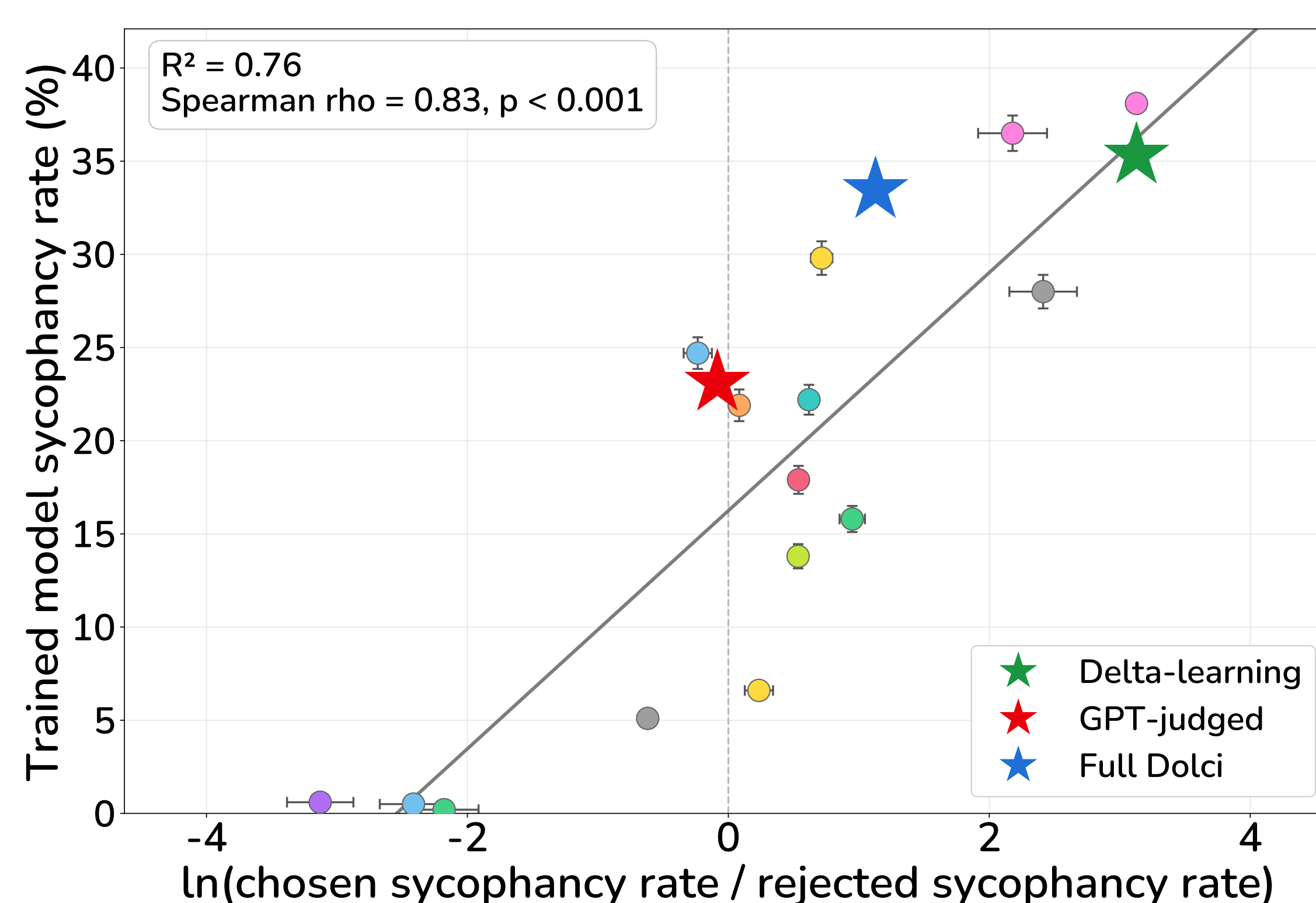


Figure 4. The teachers' log-ratio $\ln(s_C/s_R)$ predicts student sycophancy (0.6% to 35% from swapping teachers).

3 The Signal Is Diffused Across the Data

- No overt examples of sycophantic agreement
- Filtering fails:** removing 60k datapoints via probe-based attribution is ineffective, and Logit-Linear Selection subsets selected to most strongly encode sycophancy do not meaningfully increase sycophancy.

Discussion

- Teacher choice can transmit sycophancy through neutral-looking preference data.
- Actionable takeaway for model developers:** use chosen and rejected models with similar sycophancy levels.
- Limitations:** OLMo and Tulu are the only fully open-source post-training pipelines, 7–8B students, one sycophancy setting.



Paper



Contact